LinkedIn Job Postings Analytics Using PostgreSQL

1. Introduction

The purpose of this project is to design and analyze a relational database built from LinkedIn job postings data.
The project simulates a real-world analytics task where large volumes of semi-structured data must be cleaned, validated, stored efficiently, and queried to extract business insights.

PostgreSQL was chosen as the primary database system due to its advanced features such as partitioning, window functions, and robust constraint handling.

2. Data Description

The dataset consists of job postings scraped from LinkedIn and stored in CSV format.
Each posting includes:

- Job and company identifiers
- Job title and description
- Salary information (min, max, median)
- Location and remote availability
- Experience level
- Posting and expiration timestamps

Due to the size and variability of the data, careful preprocessing and validation were required before insertion.

3. Database Design

The database schema follows a normalized relational design:

- The `companies` table stores company-level information.
- The `jobs` table stores job postings and references `companies` via foreign keys.

To optimize performance, the `jobs` table is **range-partitioned by `listed_at` (posting date)**.
This allows PostgreSQL to scan only relevant partitions when querying specific time ranges, significantly improving performance for analytical queries.

4. Data Ingestion Process

Data ingestion was implemented using Python with Pandas and psycopg2.
Key challenges addressed:

- Handling invalid numeric values and bigint overflows
- Resolving foreign key mismatches

- Managing NULL values and missing timestamps
- Avoiding transaction failures via incremental commits

Rows were inserted iteratively with controlled commit intervals to ensure reliability and recoverability.

5. SQL Analysis & Queries

The project includes a diverse set of SQL queries demonstrating:

- Multi-table joins
- Aggregations and grouping
- Advanced filtering (`HAVING`, `BETWEEN`, `ILIKE`)
- Window functions for ranking and trend analysis
- CTEs for complex logic
- Date-based analytics using `DATE_TRUNC` and `EXTRACT`
- Partition-aware queries

At least two queries were analyzed using `EXPLAIN ANALYZE` to evaluate execution plans and confirm partition pruning.

6. Business Insights

The analysis provides actionable insights such as:

- Identifying the most active hiring companies
- Understanding salary distributions by experience level
- Observing hiring trends over time
- Evaluating the impact of remote work on compensation
- Detecting high-demand roles and skills

These insights can support strategic decisions in recruitment, labor market analysis, and compensation benchmarking.

7. Conclusion

This project demonstrates a full data pipeline: from raw CSV ingestion to advanced SQL analytics.
It highlights best practices in PostgreSQL database design, partitioning strategies, query optimization, and analytical SQL.

Overall, the project showcases practical skills relevant to data engineering and data analytics roles.